MOHAMED ALFATEH SHABAN

Automation — PLC — SCADA Engineer

Cairo, Egypt — Open to Relocation — +20 106 494 5394

mohamed.alfateh@ejust.edu.eg — linkedin.com/in/fateh2001

PROFESSIONAL SUMMARY

Automation, PLC & SCADA Engineer with nearly 2 years of hands-on industrial experience in **PLC programming, commissioning, troubleshooting, SCADA development, industrial networking, instrumentation, and process automation**. Strong practical experience with **Siemens S7-1200/1500, TIA Portal, WinCC Unified/Advanced, PROFINET, Modbus RTU/TCP, RS-485, VFDs, field instrumentation, FAT/SAT, and live plant commissioning**. Additional project exposure to **Siemens S7-400, STEP 7 Classic and WinCC** for industrial building management applications.
Experienced in automated **mixing, pasteurization, CIP, pumping, utilities, continuous production, and batch-process systems**. Previous DCS/SIS exposure using **Yokogawa CENTUM VP** with interlock, cause-and-effect, SIS and ESD concepts.
Seeking a Junior/Intermediate Automation Engineering position involving **PLC/DCS systems, commissioning, troubleshooting, industrial networking and control-system integration**.

CORE TECHNICAL SKILLS

PLC Platforms	Siemens S7-1200, S7-1500; project exposure to S7-400 / STEP 7 Classic
PLC Engineering	TIA Portal, STEP 7, Ladder Logic, STL, sequential control, interlocks, permissives, motor/valve control, equipment logic, I/O mapping
HMI / SCADA	WinCC Unified, WinCC Advanced, WinCC; alarms, trends, recipes, reporting, faceplates, process visualization
DCS / Safety Systems	Yokogawa CENTUM VP; exposure to SIS, ESD, cause-and-effect and process interlocking
Industrial Networks	PROFINET, PROFIBUS concepts, Modbus RTU, Modbus TCP, RS-485, OPC UA, Industrial Ethernet
Commissioning	FAT, SAT, cold/hot commissioning, loop checking, I/O validation, functional testing, startup support, fault diagnosis
Instrumentation	Level, pressure, temperature, flow transmitters; load cells; analog 4–20 mA; PT100; field-device diagnostics
Drives / Motors	VFD commissioning and parameterization, DOL, star-delta, motor protection and MCC interfaces
Standards / Concepts	ISA-88, alarm management, industrial safety concepts, P&ID interpretation, batch control
Additional PLC Exposure	Allen-Bradley, Schneider, Omron and Delta through PLC programming training
Programming / Tools	Python, Embedded C, HTML, CSS, JavaScript, AutoCAD Electrical, SolidWorks, MATLAB/Simulink

PROFESSIONAL EXPERIENCE

Automation Engineer — *Feb 2025 – Present*
Friday Ice-Cream Manufacturing — *10th of Ramadan, Egypt*

- Programmed and commissioned Siemens **S7-1200 and S7-1500 PLC systems** for mixing, pasteurization, CIP, pumping and utility processes in an operating manufacturing facility.
- Developed PLC sequences, equipment control logic, interlocks, permissives, alarms and fault handling based on P&IDs and process operating requirements.
- Developed and modified **WinCC Unified and WinCC Advanced SCADA** applications including process graphics, equipment faceplates, alarms, trends, recipes and operator monitoring functions.
- Performed **PLC and SCADA troubleshooting during live production**, diagnosing field I/O, network, communication, instrumentation, VFD and control-logic faults.
- Integrated field equipment over **PROFINET, Modbus RTU/TCP and RS-485**, including VFDs, transmitters, load cells and intelligent field devices.
- Configured and diagnosed analog instrumentation including **4–20 mA level, pressure, flow and temperature measurement systems**.
- Supported **cold and hot commissioning**, I/O checks, loop testing, functional testing, startup and production validation.
- Participated in **FAT and SAT**, interlock verification, fault testing, operator training and system handover.
- Applied **ISA-88 batch-control concepts** to process sequencing and equipment organization for repeatable production operations.
- Worked with VFD-controlled pumps, agitators and process equipment, including parameterization, control integration and fault diagnostics.

Automation Engineer – Industrial Automation / IBMS Project — Jun 2026 - Aug 2026
Freelance Engineering Project — *Remote / Singapore*

- Worked on an Industrial Building Management System based on **Siemens S7-400, STEP 7 V5.7 and WinCC 8.0**.

- Performed PLC **I/O mapping and database integration** for physical I/O and Modbus-connected building systems.
- Worked with existing STEP 7 PLC architecture and SCADA databases for electrical, fire-protection, lift, escalator, shutter and other building systems.
- Mapped PLC data to WinCC SCADA and supporting verification of commands, statuses, alarms and equipment visualization.
- Worked with **Modbus RS-485 communication** and Siemens classic PLC structures within an existing industrial automation system.

Automation Engineer *Oct 2024 – Jan 2025*
RP Minerals *Borg El Arab, Egypt*

- Maintained PLC and SCADA systems controlling **Hosokawa Alpine continuous-process production lines**.
- Troubleshot automation, electrical and communication faults during production and supported corrective maintenance to minimize downtime.
- Worked in a continuous industrial process environment requiring rapid fault diagnosis and safe restoration of production equipment.

Automation & DCS Trainee *Aug 2024 – Sep 2024*
Global Technical Services (GTS) *Maadi, Egypt*

- Worked with **Yokogawa CENTUM VP DCS** architecture and process-control engineering concepts.
- Developed and reviewed process **interlocks and cause-and-effect logic** for industrial control applications.
- Gained practical exposure to **Safety Instrumented Systems (SIS)** and **Emergency Shutdown Systems (ESD)**, including configuration and functional testing concepts.
- Worked with industrial instrumentation, control loops and process-safety engineering principles relevant to process industries.

Production & Maintenance Trainee *Aug 2023 – Sep 2023*
El-Welely Group *New Borg El Arab, Egypt*

- Supported troubleshooting of PLC-controlled production machinery and industrial electrical equipment.
- Participated in migration of relay-based control systems to **PLC/HMI-based automation**, including I/O identification, wiring modifications and commissioning support.

SELECTED AUTOMATION PROJECTS

Mixing, Pasteurization & CIP Automation – Siemens S7 / WinCC *2025–2026*

- PLC/SCADA engineering for automated food-process systems involving tanks, pumps, agitators, valves, VFDs, instrumentation and batch sequences.
- Implemented equipment interlocks, permissives, alarm handling, CIP sequencing and process visualization.
- Supported commissioning from I/O verification through live production startup and operator handover.

S7-400 / STEP 7 / WinCC IBMS Integration *2026*

- PLC/SCADA mapping for an IBMS integrating physical I/O and Modbus-connected systems using Siemens S7-400 and STEP 7 Classic.
- Worked with SCADA status, command and alarm data for multiple building-services systems.

EDUCATION

B.Sc. Mechatronics & Robotics Engineering *Jan 2025*
Egypt-Japan University of Science and Technology (E-JUST), Egypt *Grade: Very Good*

RELEVANT TRAINING & CERTIFICATIONS

- ISA-88 Batch Control Standard
- Advanced Process Control & Safety Instrumented Systems (SIS)
- Process Control & Instrumentation
- Process Visualization – Level 3
- SQL Database Programming for WinCC SCADA Engineers
- PLC Programming Training: Siemens, Allen-Bradley, Schneider, Omron and Delta
- Certified SolidWorks Professional (CSWP)

LANGUAGES

Arabic: Native **English:** Advanced **German:** A2